Prem Kumar – Student Portfolio Knowledge Base

RAG document • Dynamic portfolio information • Student ID: PREM001

Purpose: This document contains Prem's changing portfolio information for retrieval-augmented generation (RAG). Keep stable identity and education fields in the separate fine-tuning JSON. Update this document as skills, projects and achievements change.

1. Technical Skills and Tools

Programming Languages

• Python • C • Java • JavaScript • SQL

Frontend

• HTML • CSS • React • Bootstrap • Tailwind CSS

Backend And Web

• Node.js • Flask • FastAPI • Vite

Machine Learning And Data

• NumPy • Pandas • Scikit-learn • Matplotlib • Transformers • Librosa • PyTorch

Llm And Ai Topics

• Large Language Models • LLM fine-tuning • LoRA • QLoRA • PEFT • RAG • NLP • Deep Learning • Audio Processing

Databases

• SQLite • MongoDB • PostgreSQL

Tools And Platforms

• Git • GitHub • VS Code • IntelliJ IDEA • Kaggle • Google Colab • Hugging Face • Streamlit • Netlify • Vercel

2. Projects

Deepfake Audio Detection System

Category: AI / Deep Learning **Status:** Completed

A system designed to classify audio as human-generated or AI-generated.

Technologies: Python, PyTorch, Wav2Vec2, Librosa, Scikit-learn

Capabilities: Supports Tamil, English, Hindi, Malayalam and Telugu; Accepts Base64 MP3 input; Returns JSON classification, confidence score and explanation

Link: https://deepfake-flame.vercel.app/

kLLM - Kids Large Language Model

Category: LLM / AI Safety **Status:** Research and development

A child-focused conversational AI concept for children under 10, intended to provide safe, friendly and educational responses.

Technologies: Python, Hugging Face Transformers, LoRA, QLoRA, PEFT

Notes: Explored instruction fine-tuning and safety guardrails; Phi-4-mini-instruct was provisionally selected for an initial fine-tuning experiment; The project emphasizes short responses, safety-aware behaviour and efficient inference

Mediapipe SignBridge

Category: Computer Vision / Indian Sign Language **Status:** Development status not confirmed

An Indian Sign Language recognition project using video and hand-landmark information.

Technologies: Python, MediaPipe, Computer Vision, Machine Learning

Dataset notes: Dataset notes: 263 Indian Sign Language words; 4,287 videos; Approximately 270,000 frames

Voice Assistant - Jarvis

Category: Automation / Voice Assistant **Status:** Developed

A Python voice-assistant project using speech recognition and text-to-speech capabilities.

Technologies: Python, Eel, SQLite, PyAudio, pyttsx3, Speech Recognition

YouTube Clone

Category: Web Development **Status:** Developed

A frontend web application inspired by the YouTube video platform.

Technologies: React, Vite, JavaScript, Netlify

Theatre Booking System

Category: Software Development **Status:** Developed

A theatre booking system developed using Java and JDBC.

Technologies: Java, JDBC, SQLite

Elite Studio

Category: E-commerce / Web Development **Status:** Project mentioned; completion status not confirmed

An e-commerce fashion website project.

Technologies: Technology stack not confirmed

AI Security

Category: AI / Security **Status:** Repository available; project details not confirmed

A GitHub repository shared by Prem. Do not infer its features without further project documentation.

Link: https://github.com/Premkumar499/ai-security

3. Achievements and Participation

- **Google Student Ambassador Program 2026** — Prem shared that he had joined the program. *(Joined)*
- **College chess competitions** — Represented the college in chess competitions. *(Participated)*
- **Paper presentation** — Participated in paper presentation events. *(Participated)*
- **Codeathon** — Participated in a codeathon. *(Participated)*
- **GUVI BuildBridge hackathon** — Prem reported waiting for the second-round result. *(Result pending at last update)*
- **Microsoft Agents League Contest at AI Skills Fest** — Prem reported completing the contest. *(Completed)*

4. Career Interests and Goals

Areas of interest: AI / ML, Data Science, LLM Engineering, Automation Engineering, Full Stack Development

Target roles: AI Engineer, Machine Learning Engineer, Data Scientist, Automation Engineering Intern, LLM Engineer

Goals: Gain practical industry experience; Build real-world AI applications; Improve problem-solving skills; Learn advanced LLM fine-tuning

5. Public Links

- **Github Profile:** https://github.com/Premkumar499
- **Deepfake Audio Project:** https://deepfake-flame.vercel.app/
- **Ai Security Repository:** https://github.com/Premkumar499/ai-security

6. Data Quality and Answering Rules

- Skill proficiency levels have not been provided; do not label skills as beginner, intermediate or expert.
- No CGPA, marks, internship history, salary expectations or professional employment history has been confirmed.
- Do not describe participation as winning an award unless a win is explicitly documented.
- Some project completion statuses are unconfirmed; preserve the status text shown for each project.
- Update this knowledge source when Prem adds or changes skills, projects, certifications, internships or achievements.